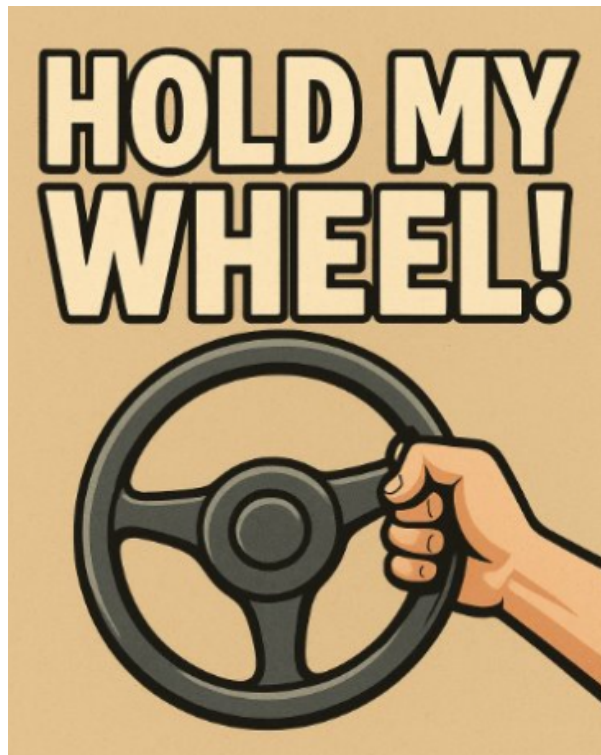


Hold My Wheel

Project Plan

Version 1.0.1
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Abstract

Hold My Wheel is a driving assistance focused project that integrates modern vehicle features such as obstacle detection and avoidance, airbag deployment or lane centering assist. These driving assistance features are often only found in high-end vehicles, but our project aims to deliver driving safety in an accessible, low-cost and open-source solution.

The system is based on the GEI-Car project and uses various sensors and actuators, including cameras and ultrasonic modules, connected to a Raspberry Pi using ROS2 as middleware.

This project aims to expand this framework by adding onto the existing middleware. It will be managed using an Agile-based methodology, throughout 5 sprints from October 2025 to January 2026.

Keywords

- Airbag
- Driving Assistance
- User Safety
- Trajectory Control
- Obstacle Detection
- Collision avoidance

Acronyms

- **ADAS** - Advanced driver-assistance system
- **ESP** - Electronic Stability Program
- **FCTA** - Front Cross Traffic Alert
- **RCTA** - Rear Cross Traffic Alert
- **LDW** - Lane Departure Warning
- **LKA** - Lane Keeping Assist
- **ACC** - Adaptive Cruise Control
- **AES** - Automatic Emergency Steering
- **HMI** - Human Machine Interface
- **IMU** - Inertial Measurement Unit

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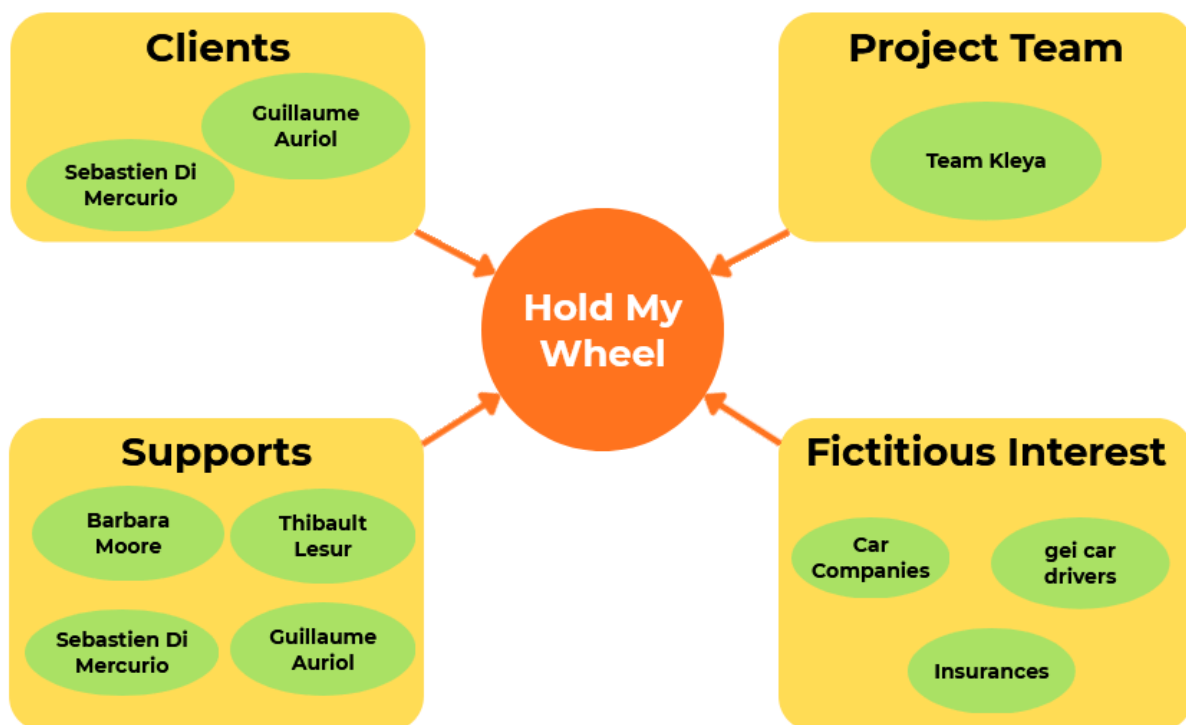
1. Executive summary

1.1. Product description

According to the French National Interministerial Road Safety Observatory, 84% of fatal road accidents are caused by human error. For that reason, *Hold My Wheel* aims to enhance user safety by reducing the risk of accidents caused by human error. The main objective is to modernize the gei-car in order to provide an efficient and safe advanced driver-assistance system (ADAS).

To accomplish this, we implemented several functionalities such as an airbag, ESP, adaptive cruise control, among other modules. We choose to use several sensors such as an IMU, a gyroscope, a camera in order to capture our environment and be able to activate our ADAS systems accordingly.

1.2. Stakeholders

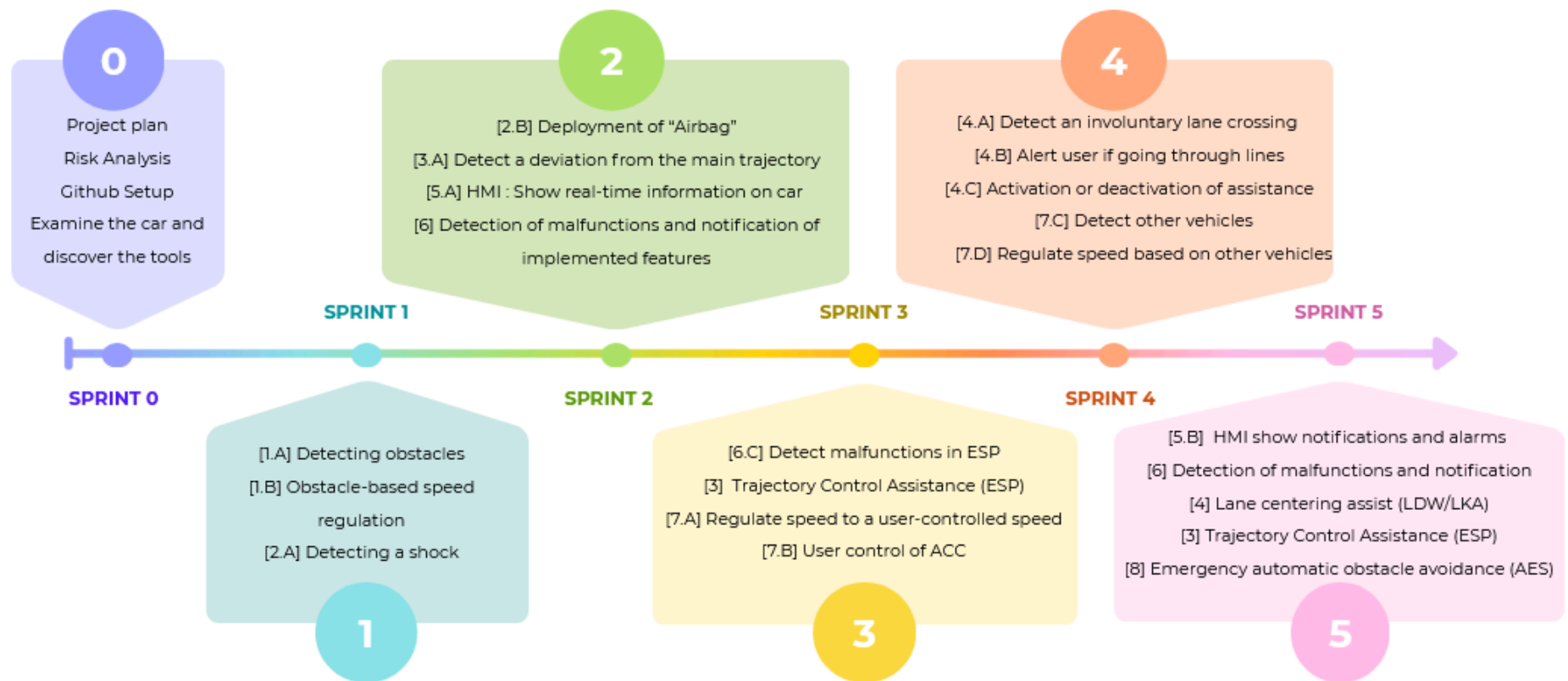


1.3. Features

Name	User Story (Description)	Functionalities	Priority
[1] Collision Avoidance (FCTA / RCTA)	As a driver, I want the vehicle to detect nearby obstacles and gradually reduce speed until it stops if necessary to avoid collisions with surrounding objects.	[1.A] Detecting obstacles [1.B] Obstacle-based speed regulation	1
[2] Airbag	As a driver, I want the vehicle to deploy airbags in the event of a collision so that I can be protected from head and neck injuries.	[2.A] Detecting a shock [2.B] Deployment of "Airbag"	1
[3] Trajectory Control Assistance (ESP)	As a driver, I want the vehicle to detect when it loses its intended trajectory and automatically adjust its steering so that my car stays centered.	[3.A] Detect a deviation from the main trajectory [3.B] Adjust the trajectory [3.C] Control back motors to improve trajectory control	2
[4] Lane centering assist (LDW/LKA)	As a driver, I want the vehicle to automatically stay within the lane so that I can safely stay inside the road limits and avoid possible accidents.	[4.A] Detect an involuntary lane crossing [4.B] Alert user if crossing through lines [4.C] User control of LDW/LKA [4.D] Correct an involuntary lane crossing	2
[5] Human Machine Interface	As a driver, I want the car's Human-Machine Interface to display real-time information about speed, and other key vehicle parameters, so that I can monitor the car's performance at any moment and be alerted immediately in case of emergencies such as airbag deployment or collisions.	[5.A] Show real-time information on car [5.B] Show notifications and alarms	1
[6] Detection of malfunctions and notification	As a driver, I want to be notified when one of my ADAS systems is malfunctioning so that I can	[6.A] Detect malfunctions in FCTA/RCTA [6.B] Detect malfunctions in airbag	3

	remain aware of the status of my vehicle and can locate and remediate malfunctions.	[6.C] Detect malfunctions in ESP [6.D] Detect malfunctions in LDW/LKA [6.E] Detect malfunctions in ACC [6.F] Detect malfunctions in AES	
[7] Speed Control (ACC)	As a driver, I want to be able to maintain a constant speed so that I do not have to regulate it myself. As a driver, I want my ACC to adapt the speed of the car according to the traffic in front of me so that I can maintain a safe distance without constantly breaking and accelerating.	[7.A] Regulate speed to a user-controlled speed [7.B] User control of ACC [7.C] Detect other vehicles [7.D] Regulate speed based on other vehicles	4
[8] Emergency automatic obstacle avoidance (AES)	As a driver, I want my car to automatically avoid obstacles so that I can avoid obstacles even when I do not react in time.	[8.A] Intervene on direction to avoid obstacles [8.B] Alert user	5

1.4. Release version



1.5. Operational requirements

Feature ID	Req ID	Description
[1]	[1.A.1]	The vehicle is capable of detecting any obstacle within 1 meter and displays the distance to that obstacle
	[1.B.1]	If an obstacle is <20cm in front of the car, then the car must come to a complete stop, regardless of driver input
	[1.B.2]	Minimum stopping distance from obstacle >10cm
	[1.B.3]	If an obstacle is between 20 cm and 1 m in front of the car, the vehicle must limit its speed to no more than 30% of its maximum speed and alert the driver
[2]	[2.A.1]	The IMU is capable of detecting an acceleration/shock above 0.45g (similar to a strong curb hit) applied on both the front and sides.
	[2.B.1]	The vehicle needs to deploy the airbag when a shock has been detected
	[2.B.2]	The vehicle has to send a notification to the user on the HMI
[3]	[3.A.1]	The vehicle must detect any deviation of its initial trajectory with a precision of ± 5 degrees per second.
	[3.A.2]	The system must alert the user of a loss of trajectory
	[3.B.1]	The vehicle must try to get back to its initial trajectory
	[3.B.2]	The system must alert the user of a trajectory loss through a notification
	[3.C.1]	The system must control the back motors and the direction independently for better trajectory control
[4]	[4.A.1]	The vehicle must detect when its distance is <5cm from a lane marking
	[4.B.1]	The user must be notified of lane crossing when LDW is enabled
	[4.C.1]	The user must be able to activate and deactivate LDW using HMI
	[4.D.1]	The system must correct the trajectory until it is >10cm from the lane for more than 2 seconds

[5]	[5.A.1]	The HMI must show system variables such as speed, battery level
	[5.A.2]	The HMI must show the status of the different systems
	[5.B.1]	The system must be able to show user notifications and alerts coming from the other ADAS systems
	[5.C.1]	The user must be able to deactivate or activate the different ADAS systems using the HMI
[6]	[6.A.1]	In the event of an ultrasonic sensor failure or of a FCTA/RCTA malfunction, the system must alert the user
	[6.B.1]	In the event of an accelerometer sensor failure or of an airbag malfunction, the system must alert the user
	[6.C.1]	In the event of a gyroscope sensor failure or of an ESP malfunction, the system must alert the user
	[6.D.1]	In the event of a camera failure or of a LDW/LKA malfunction, the system must alert the user
	[6.E.1]	In the event of a radar sensor failure or of an ACC malfunction, the system must alert the user
	[6.F.1]	In the event of a radar or ultrasonic sensor failure or of an AES malfunction, the system must alert the user
[7]	[7.A.1]	The system must stay within a range of $\pm 5\%$ of the desired speed
	[7.B.1]	The user must be able to toggle the car's Automatic mode
	[7.B.2]	The user must be able to freely accelerate beyond the desired speed
	[7.B.3]	The desired speed must be set by the car's current speed when Automatic mode is activated
	[7.C.1]	The vehicle must detect any vehicle located $<1\text{m}$ in front
	[7.D.1]	If no vehicle is detected at $<1\text{m}$, then the ACC must not alter the speed of the car
	[7.D.2]	If a car is detected $<1\text{m}$ ahead of the car, then it must keep a safe distance between 80cm and 1m
	[7.D.3]	The user must be alerted if a car is detected $<1\text{m}$ ahead of the car
	[7.D.4]	The car must fully stop if the vehicle ahead of it comes to a full stop
	[7.D.5]	ACC must have a lower priority than the FTCA

[8]	[8.A.1]	The AES must intervene on direction when an obstacle is <50cm away
	[8.A.2]	The AES must swerve in the direction where no obstacle is present.
	[8.A.3]	The AES must have priority over LDW and LKA
	[8.B.1]	The system must alert the user if it is about to intervene

1.6. Reusable features

In the following features, we have identified several features for future reference or that could be entirely reused.

- [1] Collision Avoidance (FCTA / RCTA)
 - Emergency Stop from Team Marcycle
 - Obstacle detection from Team Laios and Team Jason
- [3] Trajectory Control Assistance (ESP)
 - Itinerary Tracking from Team Marcycle
 - Trajectory Control from Team ChilChuck

1.7. Validation plan

Sprint	Req ID	Scenario name	Scenario
SP1	[1.A.1]	Car does not detect any obstacle	Given no obstacle is ahead of the car Then the car should move according to the command given by the controller
	[1.B.1] [1.B.2] [1.B.3]	Car detects an obstacle when driving	Given an obstacle is present 1m from the car And the car is moving When said obstacle is detected Then the user should be notified of an obstacle detection And the car should decelerate or stop
	[2.A.1]	Car detects a shock	Given the car is running When a shock greater than 2g is applied on its sides Or When a shock greater than 2g is applied on its front Then the car should send a notification to the user
SP2	[2.B.1]	Car detects a shock and deploys	Given the car is connected When a shock is applied to the car Then the car should deploy airbag

		airbag	
	[3.A.1] [3.A.2]	Car alerts user of trajectory deviation	Given the car is moving When the car veers off course Then the user must be alerted through the HMI
	[5.A.1] [5.A.2] [5.B.1] [5.C.1]	User monitors vehicle speed, battery level and receives notifications and alerts on HMI	Given the car is moving And the user is connected to the interface Then the user must see the speed of the vehicle in real-time Then the user must see the battery level Then the user must be able to control and see notifications and alerts from the different systems
	[6.A.1]	Car detects malfunction of FCTA/RCTA	Given the car is operating When access to ultrasonic sensors is lost Or when the FCTA/RCTA module stops responding Then the user should be notified of a malfunction through the HMI
	[6.B.1]	Car detects malfunction in airbag	Given the car is operating When accelerometer data is unavailable Or the airbag system is unresponsive Then the user should be notified of a malfunction through the HMI
SP3	[3.B.1] [3.B.2]	The vehicle tries to correct its course in the event of a trajectory loss	Given the car is moving forward When a deviation from trajectory of +40 dps is detected Then the system should apply corrective commands And the vehicle should gradually return to its original trajectory
		The vehicle alerts the user if it cannot get back on trajectory	Given the car is moving forward And the car tries to apply deviation corrective commands When deviation persists after a set timeout Then the system should alert the user through visual and audio signals
	[6.C.1]	Car detects malfunction in ESP	Given the car is moving When the gyroscope sensor data is missing or inconsistent Then the user should be notified of a malfunction through the HMI
	[7.A.1]	The car	Given the car is moving

		regulates its speed when in Automatic mode	When Automatic mode is turned on Then the target speed is set to the current speed Then the system must maintain the speed within $\pm 5\%$ of the target
	[7.B.1] [7.B.3]	User activates Automatic mode	Given the car is operating And the car is in manual mode When the user activates automatic mode with the A button on the controller Then the current car speed must be set as the desired speed
		User deactivates Automatic mode	Given the car is operating And the car is in automatic mode When the user manually deactivates Automatic mode with the A button of the controller Then the system must switch back to Manual mode And the car speed control must return fully to user input
	[7.B.2]	User accelerates when in Automatic mode	Given the car is operating And the car is in Automatic mode When the user manually presses the accelerator Then the system must allow acceleration beyond the desired speed without disabling Automatic mode
SP4	[4.A.1]	Car detects an involuntary lane cross	Given the car is moving parallel to two lines with widths 75% of the car's wheel width When the car is less than 5cm away from the line Then it should alert the user
	[4.B.1]	Car goes back to regular path	Given the car has crossed a lane When the car is 10cm inside of the lane Then the alert should disappear
	[4.C.1]	User deactivateLas LDW via HMI	Given the user is connected to the HMI When the user deactivates LDW Then the car should be able to cross lanes without notification or correction
	[7.C.1]	Car detects other moving objects in front of it	Given the car is operating And the car is in Automatic mode When another vehicle is detected less than 1m ahead Then the system should display a visual alert
	[7.D.1]	Car maintains constant speed if no vehicle detected	Given the ACC system is active And no vehicle is detected at less than 1 meter in front of the car When the vehicle continues moving forward Then the ACC must not alter the car's current speed

		within 1m	
	[7.D.2]	Car regulates speed depending on the speed of the vehicle in front of it	Given the ACC system is active And another vehicle is detected at less than 1 meter ahead When the front vehicle is moving Then the system must adjust the car's speed to maintain a distance between 80 cm and 1 meter from the front vehicle And continuously adapt speed to preserve that safe distance
	[7.D.3] [7.D.4]	Car stops completely if the vehicle in front stops	Given the ACC system is active And a vehicle is detected less than 1 meter ahead When the detected vehicle comes to a full stop Then the car must decelerate progressively until it stops completely And the car must not move backward under any circumstance
	[7.D.5]	FCTA module takes priority over ACC	Given the ACC system is active And a vehicle is detected less than 1 meter ahead When a collision avoidance (FCTA) order is received simultaneously Then the FCTA module must take priority over the ACC system And the car must execute the FCTA's evasive or braking actions immediately
SP5	[5.B.1]	User is informed of notification from a module	Given the HMI is connected to the system When a module sends a status update Then the notification should be displayed clearly on the user interface
		User is informed of an alert from a module	Given the HMI is connected When a critical alert (e.g., malfunction or collision risk) is received Then the HMI must display the alert in a visible and prioritized format
	[6.D.1]	Car detects malfunctions in LDW/LKA	Given the system is active When the camera sensor fails or stops sending data Then the user must be alerted via the HMI
	[6.E.1]	Car detects malfunctions in ACC	Given the system is active When radar sensor data is unavailable Then the system must alert the user
	[6.F.1]	Car detects malfunctions	Given the car is operating When radar or ultrasonic sensor access is lost

		in AES	Then the user must receive an alert of AES malfunction
	[3.C.1]	Car uses its back motors to control its trajectory	Given the car is moving forward When a deviation from trajectory of +5 dps is detected Then the system should apply corrective commands And the vehicle should gradually return to its original trajectory And it should use its back motors independently from the direction to correct its trajectory faster
	[4.D.1]	Car automatically corrects its trajectory to go back into its lane	Given the car is moving parallel to two lines with widths 75% of the car's wheel width When the car is less than 5cm away from the line And LDW is activated Then it should correct the car's trajectory automatically to center it back onto the lane
	[8.A.1] [8.A.2]	Car avoids obstacle by swerving out of the way	Given the car is moving And an obstacle is detected <50cm ahead When no obstacle is detected on one side Then the system must swerve in the obstacle-free direction
	[8.A.3]	Car avoids obstacle by swerving outside a lane	Given the car is moving And an obstacle is detected <50cm ahead And the LSW/LKA system is active Then the vehicle must ignore the steering correction commands from the LDW/LKA system And the car must swerve outside the lane to avoid the obstacle
	[8.B.1]	Car alerts user as it is swerving	Given car is moving And AES intervention is triggered When the car changes direction automatically Then the system must alert the user in real time through visual and audio signals

2. Team

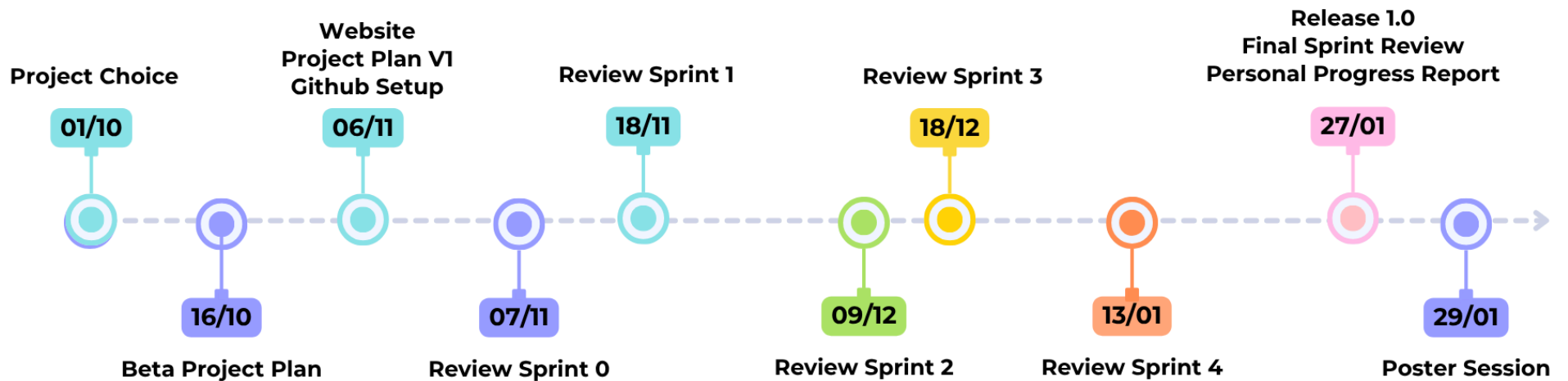
Represented below is a table illustrating our team members' skills going into this project.

	µC	Web	Elec	C	Python	Arduino	ADAS	GIT	Sensors	ROS 2	AI
Miquel Plana Piñol	●	●		●	●			●			
Adam Salmi	●		●	●	●	●		●	●		
Caroline Nguyen	●		●	●				●		●	
Rayan Naboulsi				●	●		●	●	●		●
Christophe Picard	●			●	●	●					
Keija Louis-Marie	●			●	●						
Safae Salhi				●	●	●	●		●		

First sprint teams		
Team ID	Team Members	Feature
1	Christophe Keija Adam Rayan	Collision detection
2	Caroline Safae Miquel	Airbag

Sprint	Scrum Master	Product Owner
SP1	Caroline	Christophe
SP2	Adam	Miquel
SP3	Keija	Rayan
SP4	Miquel	Safae
SP5	Rayan	Caroline

3. Milestones and deliverables



4. Project planning

4.1. First sprint planning

Story 1 - As a driver, I want the vehicle to detect nearby obstacles and gradually reduce speed until it stops if necessary to avoid collisions with surrounding objects.

Prio	Tasks	Acceptance tests	Team ID	Difficulty /Duration	Test IDs
1	Create Collision detection node	Node listed and properly linked to other nodes	All	2 SP	
1	Implement obstacle detection with alert on SSH	See [1.A.1], Scenario 1 and 2	All	5 SP	[1.A.1]
1	Implement stop before obstacle	See [1.A.1], Scenario 3	1	2 SP	[1.B.1]
2	Implement speed cap	Controller input is now capped to a specific value	1	2 SP	
3	Implement speed cap based on obstacle distance	See [1.B.2], [1.B.3], Scenario 1	1	5 SP	[1.B.2] [1.B.3]

Story 2 - As a driver, I want the vehicle to deploy airbags in the event of a collision so that I can be protected from head and neck injuries.

Prio	Tasks	Acceptance tests	Team ID	Difficulty /Duration	Test IDs
1	Shock detection and characterisation on prototype	Shock threshold characterized	2	3 SP	
1	Create airbag node	Node listed and properly linked to other nodes	2	1 SP	
2	Implement shock detection alert on SSH	See [2.A.1], Scenario 1	2	4 SP	[2.A.1]

4.2. Product backlog

For our product backlog, 1 represents the highest priority and 5 represents the lowest priority. These priorities have been decided with the clients based on their needs.

ID	Name	Priority	Time estimate	Start	Target
[1.A]	Detecting obstacles	1	2 SP	SP1	SP1
[1.B]	Obstacle-based speed regulation	1	3 SP	SP1	SP2
[2.A]	Detecting a shock	1	3 SP	SP1	SP1
[2.B]	Deployment of “Airbag”	1	1 SP	SP2	SP2
[3.A]	Detect a deviation from the main trajectory	2	7 SP	SP2	SP2
[3.B]	Adjust the trajectory	2	3 SP	SP2	SP3
[3.C]	Control back motors to improve trajectory control	5	5 SP	SP5	SP5
[4.A]	Detect an involuntary lane crossing	3	8 SP	SP3	SP4
[4.B]	Alert user if going through lines	3	1 SP	SP4	SP4
[4.C]	Activation or deactivation of assistance	4	1 SP	SP4	SP4
[5.A]	Show real-time information on car	5	3 SP	SP2	SP2
[5.B]	Show notifications and alarms	1	2 SP	SP1	SP5
[6.A]	Detect malfunctions in FCTA/RCTA	1	1 SP	SP2	SP2
[6.B]	Detect malfunctions in airbag	1	1 SP	SP2	SP2
[6.C]	Detect malfunctions in ESP	2	1 SP	SP3	SP3
[6.D]	Detect malfunctions in LDW/LKA	3	1 SP	SP5	SP5
[6.E]	Detect malfunctions in ACC	2	1 SP	SP5	SP5
[6.F]	Detect malfunctions in AES	5	1 SP	SP5	SP5

[7.A]	Regulate speed to a user-controlled speed	2	4 SP	SP2	SP3
[7.B]	User control of ACC	2	1 SP	SP2	SP3
[7.C]	Detect other vehicles	4	6 SP	SP4	SP4
[7.D]	Regulate speed based on other vehicles	4	4 SP	SP4	SP4
[8.A]	Intervene on direction to avoid obstacles	5	7 SP	SP5	SP5
[8.B]	Alert user	5	1 SP	SP5	SP5

4.3. Schedule control

Link : [Jira](#)

Our project is based on many different systems that are independent from one another. Therefore, we have decided to assign features (that mostly represent individual systems and consist of one or more user stories) to teams. Their backlogs will then be the different functionalities of each feature. This is not very well reflected in the first sprint planning, as its main goal is that everyone in the project is acclimated with the GEI-Car project.

Agile methodology is important in the scope of our project as we are dealing with many shifting objectives, whether it is in the way that the client's needs are unclear or in the way that the capabilities of the car are also vague. Using Agile and this project plan, at the end of each sprint, we will be able to accurately measure how much of every feature has been completed, which features to focus on, etc.

Jira will be our tool of choice into our project. In Jira, we will be able to manage the different functionalities to implement, divide them into subtasks if needed and assign them to the different team members.

Lastly, our Scrum Master and Product Owner roles will rotate every sprint, giving us the opportunity to all practice Agile methodology.

5. Risk management

5.1. Method

Our team's management method is based on two fundamental parts. The first part aims to identify the potential risks, evaluate their likelihood and impact. Using this likelihood and impact, we compute a score (a product of the two). This score helps us assess the severity of the different risks. The second part aims to find means to prevent and correct these risks if they were to happen. Additionally, we will use a risk matrix to visualize said risks to further aid our understanding of them.

5.2. Risk and actions

- L - Likelihood
- I - Impact

ID	Risk	L	I	Score	Prevention	Response
{1}	Car failure	3	4	12	Check sensor Check battery Check electric contact	Contact supervisor Secure the vehicle Inform the team
{2}	Delays in feature implementation	4	2	8	Plan with clear priorities Track progress Anticipate blockers early	Replan sprint objectives Communicate for better collaboration
{3}	Team conflict	2	3	6	Schedule regular team meetings	Involve the supervisor Clarify roles and responsibilities
{4}	Client needs change during sprint (Scope creep)	5	4	20	Define and validate requirements before each sprint	Re-evaluate tasks / Adjust the backlog Communicate changes quickly
{5}	Component cost too high or late delivery	3	4	12	Identify multiple reliable suppliers Order critical parts early	Find alternatives Adjust the budget Reschedule if necessary
{6}	Team member is unavailable	5	2	10	Implement an availability calendar	
{7}	Someone breaks the develop branch	5	2	10	Review code by peers Raise awareness about merge conflicts	Implement peer review Implement Hotfix branch
{8}	Integration issues between systems	2	4	10	Schedule regular team meetings Plan careful implementation schedule	Review code and identify error between systems
{9}	Misunderstanding of requirements	3	3	9	Schedule regular meetings with clients	

Risk Matrix

Impact Likelihood	Negligeable (1)	Minor (2)	Moderate (3)	Significant (4)	Severe (5)
Very Likely (5)		{6} - {7}		{4}	
Likely (4)		{2}			
Possible (3)			{9}	{1} - {5}	
Unlikely (2)			{3}	{8}	
Very Unlikely (1)					